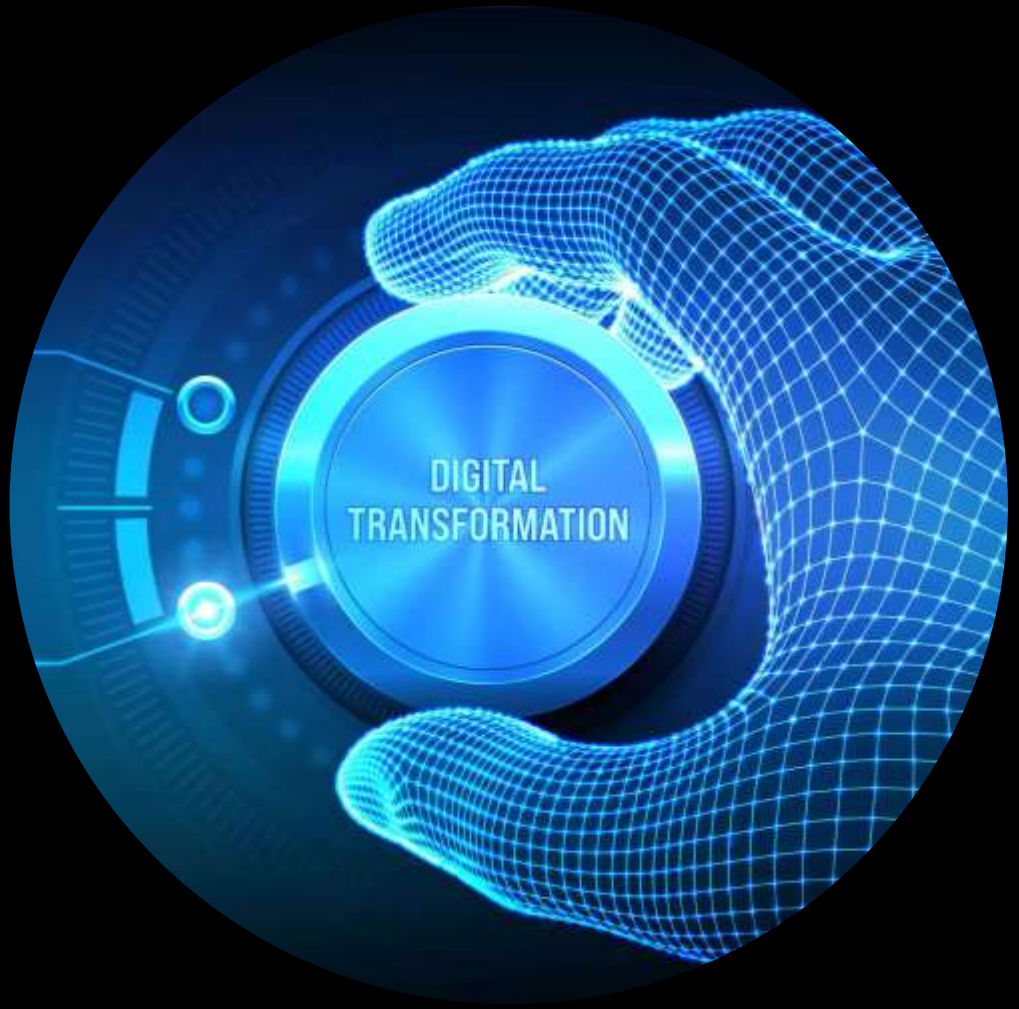


Digital transformation

Dr. Sherif Mohamed Tawfik



Objectives

- Digitization, Digitalization and Digital Transformation
- Definition
- Inventions
- Steps
- Types
- Challenges
- Benefits & Limitations
- COVID-19 pandemic changed digital transformation
- Examples
- The future

Stages of Information Technology Development



2023
Revolution of Artificial Intelligence
Digitalization



2018
Smart Applications
Smart Computer
Internet of Things



2013
Mobile Application
Smart Phone & cloud computing
Wireless Internet



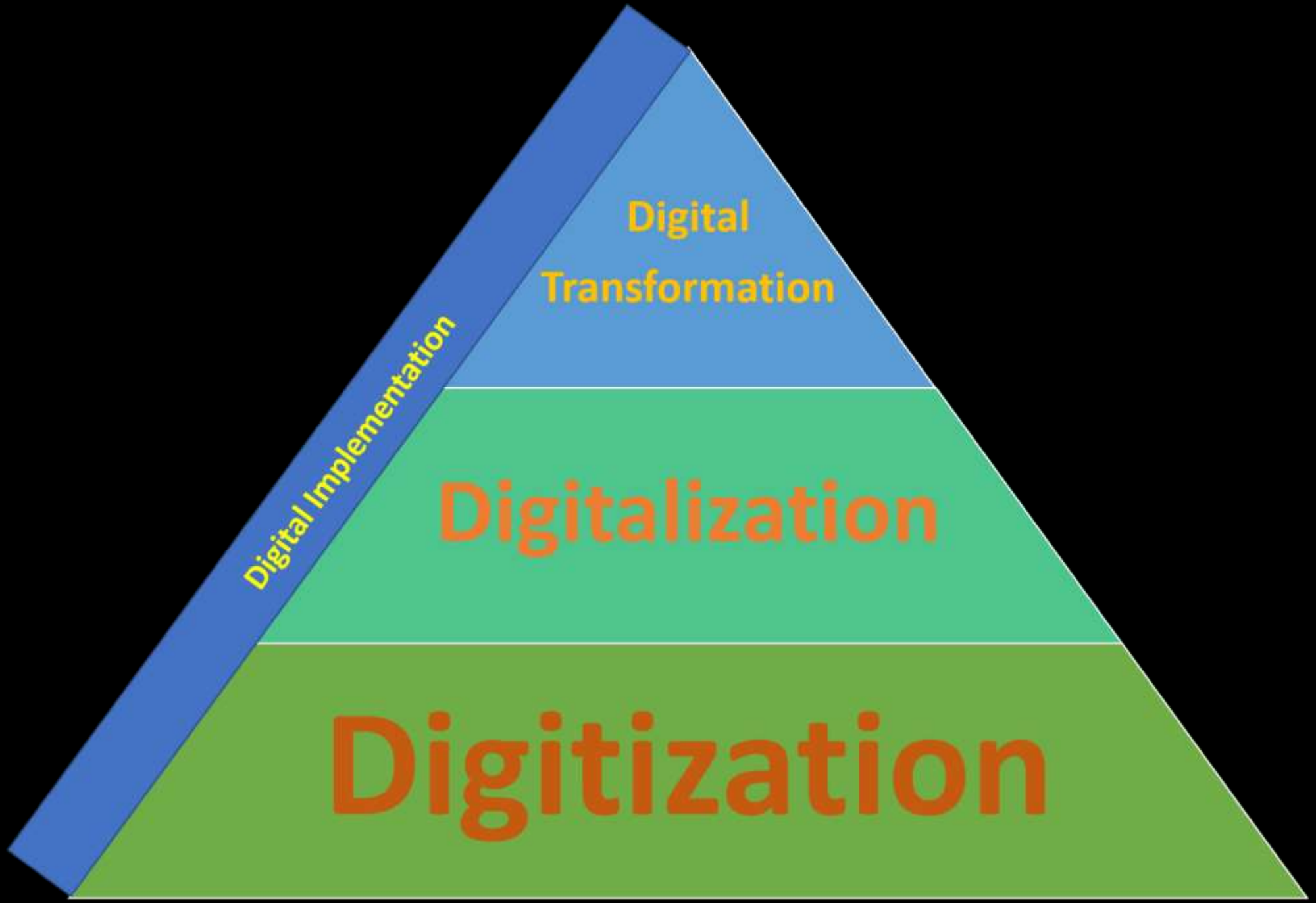
2008
Web based Applications
tablet Computer
High speed Internet



2003
Windows Applications
Laptop Computer
Medium speed Internet



1998
Text Application
Desktop Computer
Low speed Internet


Digital Implementation
Digital Transformation
Digitalization
Digitization

DIGITIZATION

Digitization is the **move from analog to digital**.

Not so long ago, businesses kept records on paper. Whether handwritten in ledgers or typed into documents, business data was analog.

Then computers went mainstream, and most businesses started converting all those ink-on-paper records to digital computer files.

DIGITALIZATION

Digitalization is **using digital data to simplify how you work**.

The process of **using digitized information to make established ways of working simpler and more efficient** is called digitalization. It’s about keeping on keeping on, but faster and better now that your data is instantly accessible and not trapped in a file cabinet somewhere in a dusty archive.

DIGITAL TRANSFORMATION

Digital transformation **adds value to every customer interaction**.

Digital transformation is changing the way business gets done and, in some cases, creating entirely new classes of businesses.

With digital transformation, companies are taking a step back and revisiting everything they do, from internal systems to customer interactions both online and in person.

What is digital transformatiOon

- The process of using digital technologies to create new or modify existing business processes ,culture ands customer experiences to meet changing business and market requirements

- Integration of digital technologies into all areas of a business resulting in fundamental changes to how business operate and how they deliver value to customer

50 Years Of Excellence



Information & Documentation Center

Information & Documentation Center

Smart Services

Operations, Training & Tech. Support

Website Development

Decision Support Systems

Mobile Applications

Workflows Automation

ERP

Digital Transformation





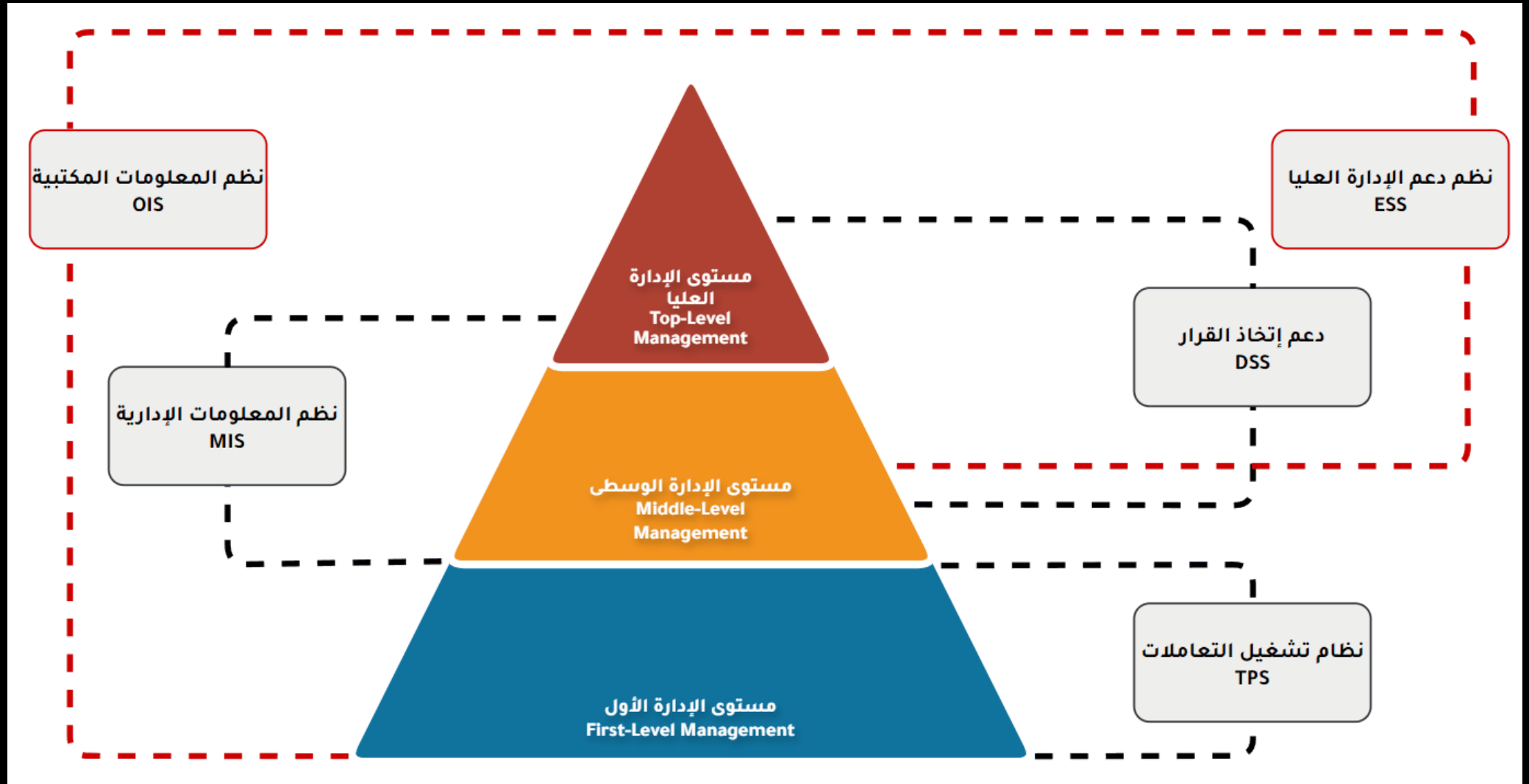


Analytics

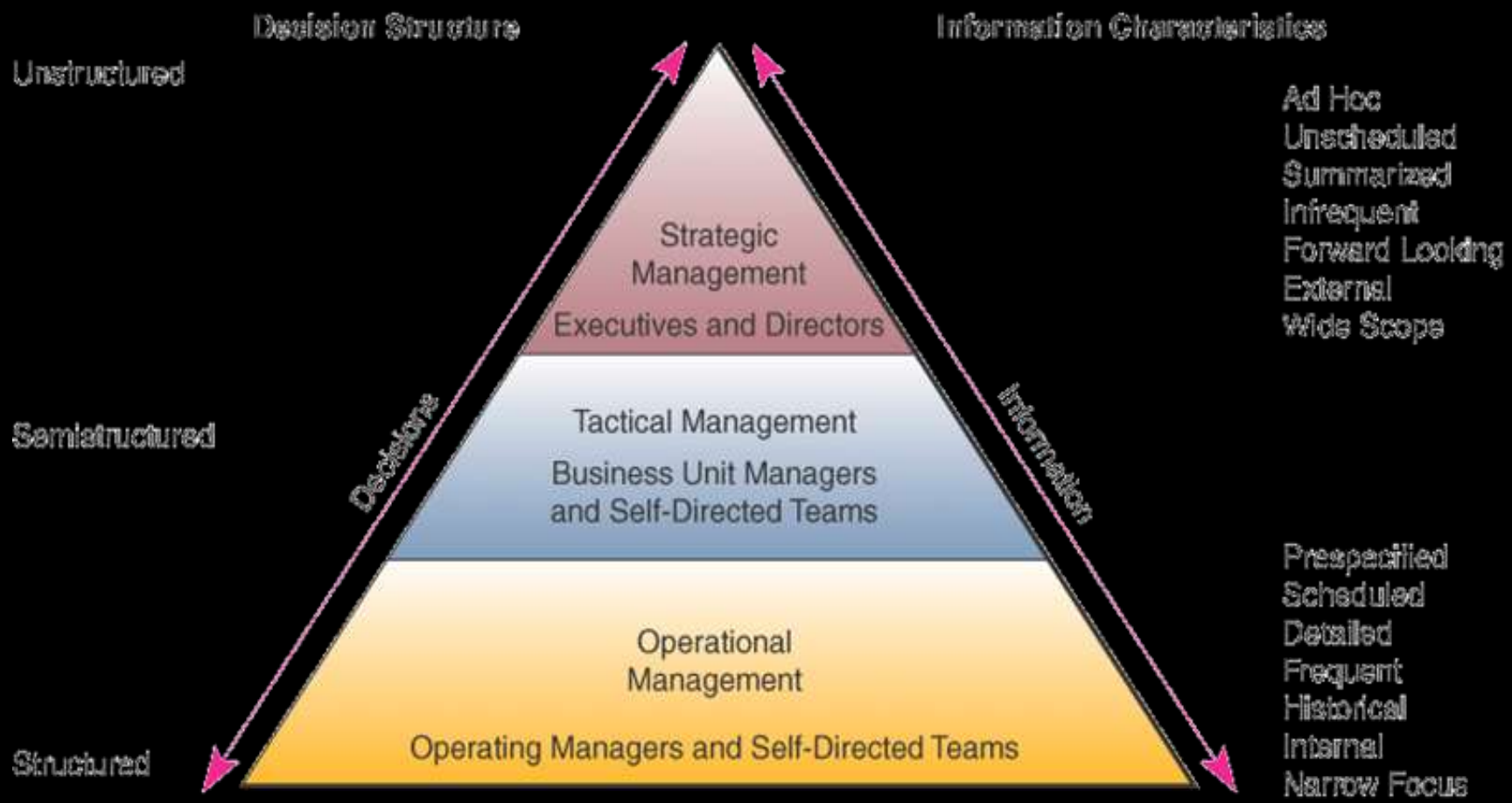
Reporting

Data Acquisition

The relationship between organizational management levels and digital transformation

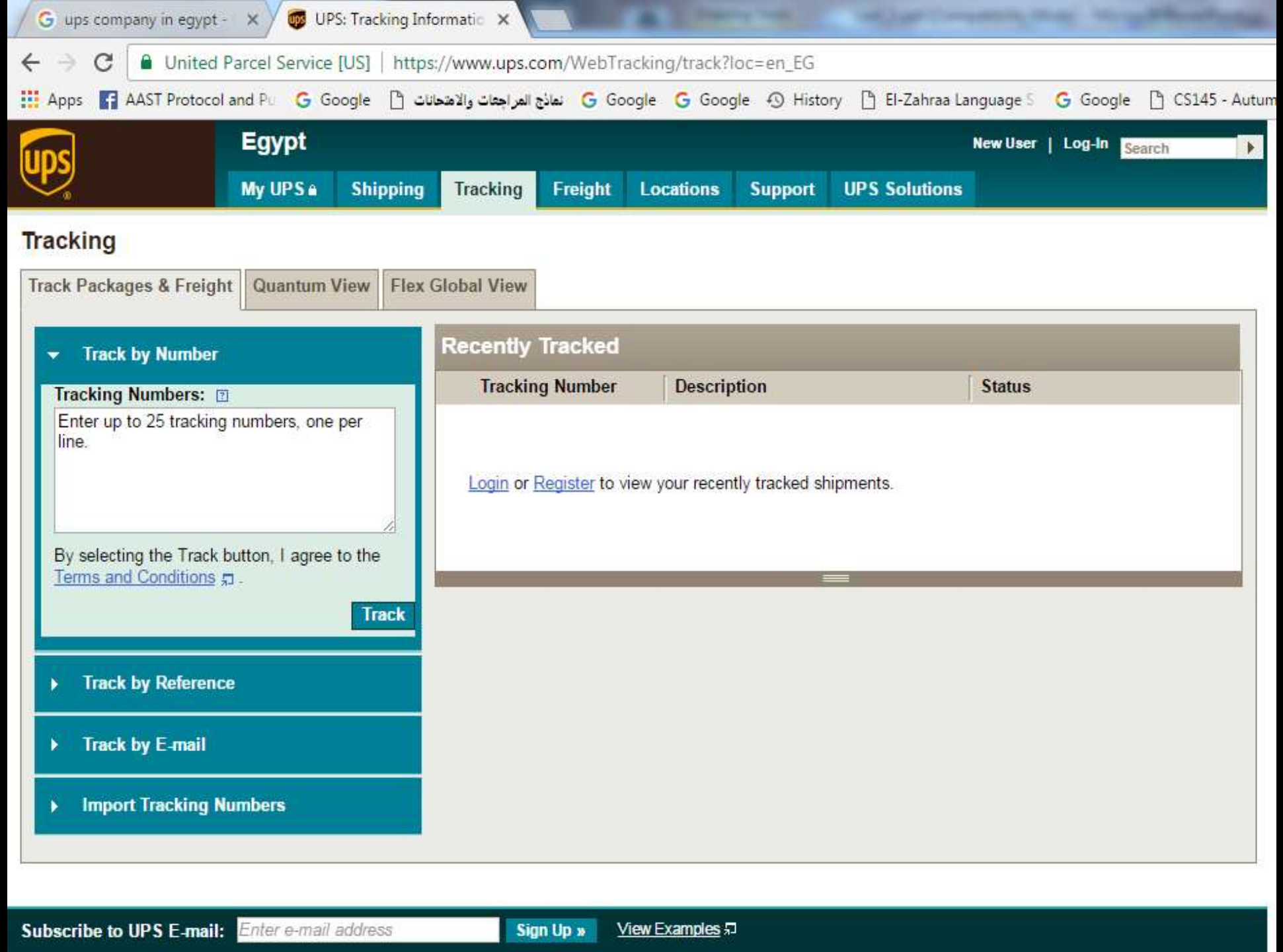


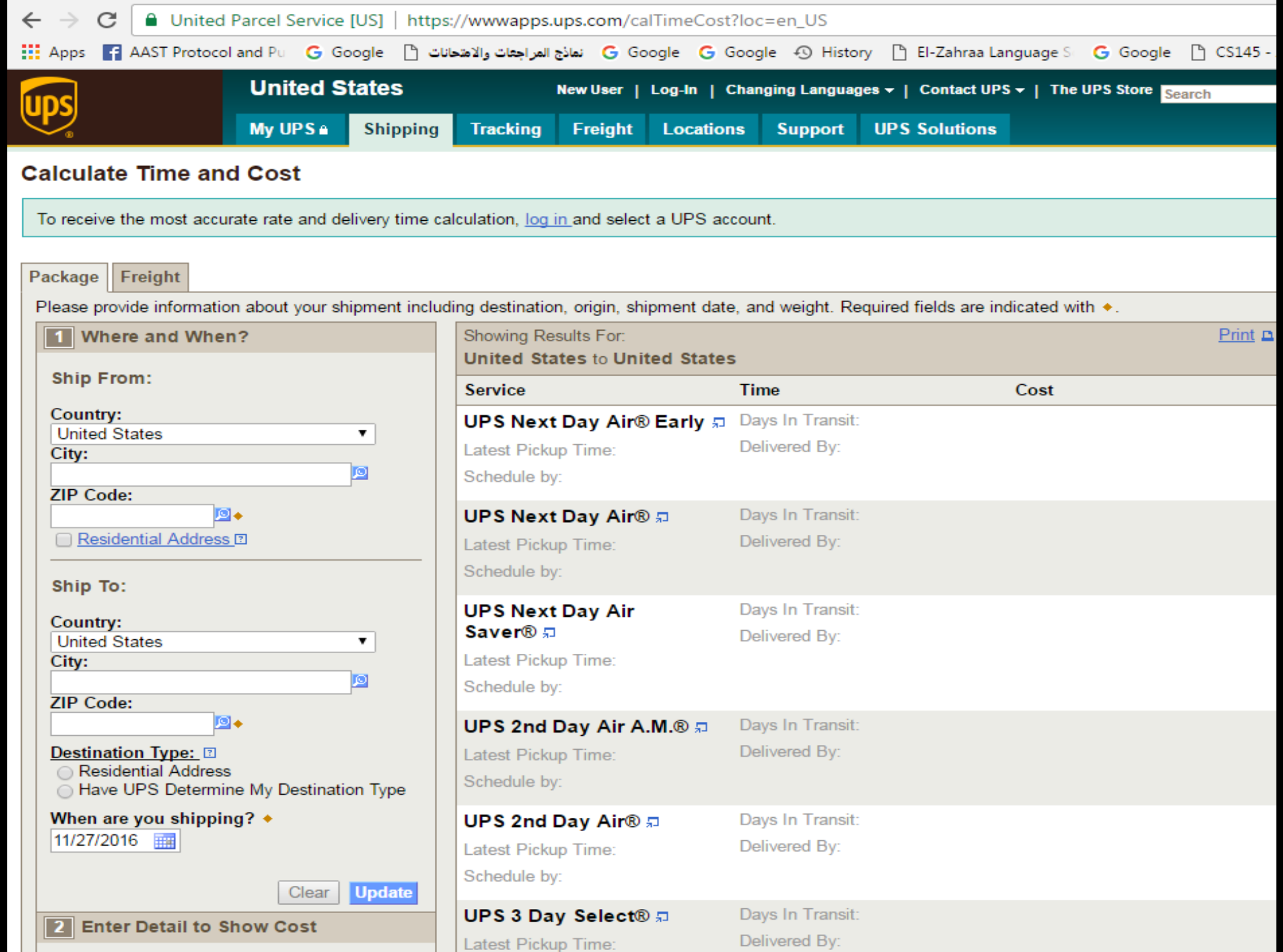
The relationship between organizational management levels and Decision making



Why are information systems so essential for running and managing a business today?

- **Businesses invest in IT to achieve six important business objectives.**
  1. **Operational excellence**
  2. **New products, services, and business models**
  3. **Customer and supplier intimacy**
  4. **Improved decision making**
  5. **Competitive advantage**
  6. **Survival**


ups company in egypt -
UPS: Tracking Informatio
United Parcel Service [US] | https://www.ups.com/WebTracking/track?loc=en_EG
Apps
AAST Protocol and Pu
Google
نماذج المراجعات والامتحانات
Google
Google
History
El-Zahraa Language S
Google
CS145 - Autum
Egypt
New User | Log-In
Search
My UPS
Shipping
Tracking
Freight
Locations
Support
UPS Solutions
Tracking
Track Packages & Freight
Quantum View
Flex Global View
Track by Number
Tracking Numbers:
Enter up to 25 tracking numbers, one per line.
By selecting the Track button, I agree to the Terms and Conditions .
Track
Track by Reference
Track by E-mail
Import Tracking Numbers
Recently Tracked
Tracking Number
Description
Status
Login or Register to view your recently tracked shipments.
Subscribe to UPS E-mail:
Enter e-mail address
Sign Up »
View Examples



United Parcel Service [US] | https://wwwapps.ups.com/calTimeCost?loc=en_US

Apps | AAST Protocol and Pu | Google | نماذج المراجعات والامتحانات | Google | Google | History | El-Zahraa Language S | Google | CS145 -

United States | New User | Log-In | Changing Languages | Contact UPS | The UPS Store | Search

My UPS | Shipping | Tracking | Freight | Locations | Support | UPS Solutions

Calculate Time and Cost

To receive the most accurate rate and delivery time calculation, log in and select a UPS account.

Package | Freight

Please provide information about your shipment including destination, origin, shipment date, and weight. Required fields are indicated with ◆.

1 Where and When?

Ship From:

Country: United States

City:

ZIP Code:

Residential Address

Ship To:

Country: United States

City:

ZIP Code:

Destination Type:
- Residential Address
- Have UPS Determine My Destination Type

When are you shipping? ◆ 11/27/2016

Clear | Update

2 Enter Detail to Show Cost

Showing Results For:
United States to **United States**

Print

Service	Time	Cost
UPS Next Day Air® Early	Days In Transit: Delivered By:	
Latest Pickup Time: Schedule by:		
UPS Next Day Air®	Days In Transit: Delivered By:	
Latest Pickup Time: Schedule by:		
UPS Next Day Air Saver®	Days In Transit: Delivered By:	
Latest Pickup Time: Schedule by:		
UPS 2nd Day Air A.M.®	Days In Transit: Delivered By:	
Latest Pickup Time: Schedule by:		
UPS 2nd Day Air®	Days In Transit: Delivered By:	
Latest Pickup Time: Schedule by:		
UPS 3 Day Select®	Days In Transit: Delivered By:	
Latest Pickup Time:		



DIGITAL TRANSFORMATION STRATEGY

7 ESSENTIAL STEPS TO DRIVE DX SUCCESS IN THE ENTERPRISE

01. ALIGN ON THE WHY

Define the business value of digital transformation.

02. PREPARE FOR CULTURE CHANGE

Put humans at the heart of digital transformation.

03. START SMALL BUT STRATEGIC

Jump-start digital transformation with an impactful, measurable initiative.

04. MAP OUT TECHNOLOGY

Choose tech solutions that scale with your long-term strategy.

05. SEEK OUT PARTNERS & EXPERTISE

Do more – and do it faster – with partners that share in your vision.

06. GATHER FEEDBACK & REFINE

Be flexible, learn from experience, and adjust accordingly.

07. SCALE & TRANSFORM

New ways to improve will emerge – embrace them!



ptc

TYPES OF DIGITAL TRANSFORMATION

1. BUSINESS PROCESS

Companies can **revise internal processes** to lower costs, improve quality and reduce cycle times.

For example, **adopting cloud connectivity** can link processes and **implementing robotic process automation** can transform manual tasks found in procurement, supply chain management and other administrative functions.

Modernizing your logistics network and supply chain by digitizing these processes, while integrating machine learning and artificial intelligence, helps to **recognize and shape data patterns into actionable insights**.

2. BUSINESS MODEL

Business model transformation aims to fundamentally **change the way companies deliver value for customers**.

For example, Netflix made the switch from mailing DVDs to online streaming, while Blockbuster failed to make the transition, which ultimately led to the latter company's eventual fate.

Technology companies can **reshape their go-to-market strategy** and support their customers' digital transformation efforts with the flexibility to select technology that best suits their needs.

3. DOMAIN

Organizations typically adopt new technologies to **redefine their products and services**.

They might extend current services to **a new customer base** or **develop entirely new technology-enabled offerings**.

For example, offering equipment on a rental basis rather than purchase-only enables you to reach a previously unserved segment of customers who may only need your technology temporarily.

4. CULTURAL/ORGANIZATIONAL

- adopt agile workflows
- develop a bias towards testing and learning
- support decentralized decision-making

However, a successful transition to a digital-first culture requires **redefining mindsets and processes** while also **incorporating new talents and capabilities**.

A typical mindset shift occurs from **managing production output** to focusing on **customer service and innovation**. Often, the cultural shift occurs organically during other transformation initiatives as internal teams adopt digital workflows and recognize the power of changing organizational norms.

5. CLOUD TRANSFORMATION

- helps organizations migrate information systems to cloud computing environments .
- for example, a company can migrate only specific applications, data, or services, and retain some legacy infrastructure, or move their entire infrastructure to the cloud.
- Cloud innovation offers many benefits, including more efficient data sharing and storage, faster time-to-market, and greater organizational scalability and flexibility.

What are the challenges in implementing a digital transformation strategy?

- Addressing inertia
- No support from leadership
- Unrealistic ambitions
- Lack of Customer Centricity
- Compliances and Roadblocks



What are the challenges in implementing a digital transformation strategy? (contd.)

1- Addressing Inertia
Within companies, there is inertia to change and if not overcome, it becomes a pushback to digital transformation. This pushback can be cultural or come from individual employees who resist change.

2- No Support from Leadership
Something as extensive and expensive as transformation needs sponsors. Usually, it is the leaders or the C-suite of the company that facilitates this change. But if the leadership team is not convinced of the need to transform, the process is ridden with challenges.

3- Unrealistic Ambitions
Digital transformation is by no means an easy task to undertake. So, if you bite more than you can chew, your plans will not go as expected. Organizations need to set doable and realistic targets for themselves while transforming.

What are the challenges in implementing a digital transformation strategy? (contd.)

4: Lack of Customer Centricity

It is important to remember that the customer is at the forefront of this change. Transformation for its own sake will not add value to the company. The process yields results only when trying to make the customer’s life easier.

5: Compliances and Roadblocks

The use of digitally transformative technologies comes with its own set of regulations and compliances. For instance, if you want to use blockchain technology, you must follow the regional and global compliances for the same. This can pose substantial challenges to organizations that are still learning the ropes.

—Tim O’Reilly | Founder & CEO of O’Reilly Media

“Every industry and every organization will have to transform itself in the next few years. What is coming at us is bigger than the original internet, and you need to understand it, get on board with it, and figure out how to transform your business”

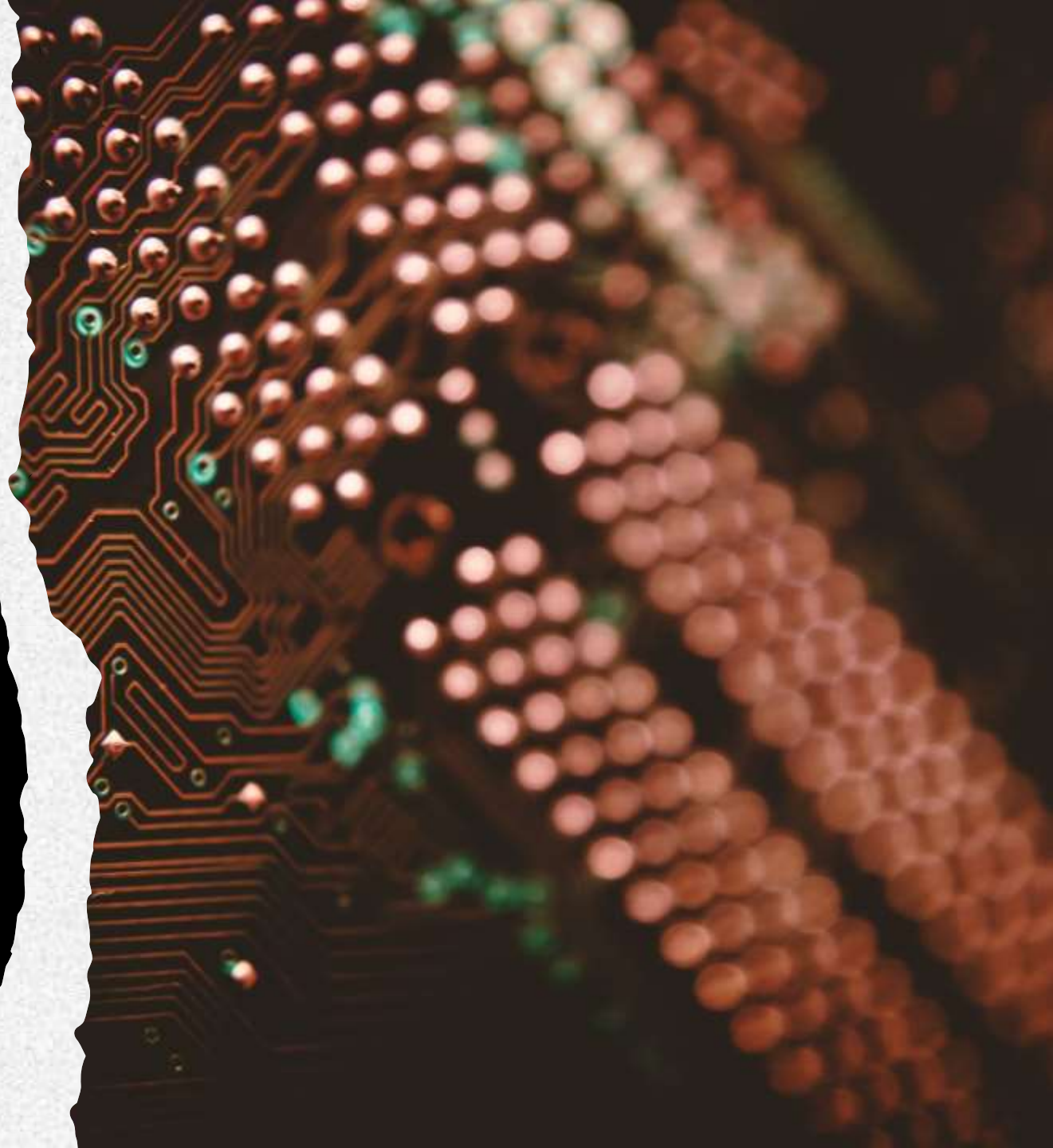

Benefits of
Digital
Transformation:

1) **Digital transformation puts the customer at the** ot sknahT :gnihtyreve fo retnec ecneirepxe gnippohs eht sevitanretla enohp-rep-yap ro sppa enohptrams ,ppAstahW .rehtooms neve emoceb nac sessecorp gnirutcafunam eht dna
2) **Real-time Data and Monitoring:** noitca ,maet ,niahc ylppus cificeps a fo eulav ehT dna devihcra atad emit-laer yb tub ,noitiutni yb dediug erom on si noisiced ro detneiro-laog dna eliga ,detcennoc erom etaerc lliw ohw ,seeyolpme yb dezinagro .smaet
3) **Meeting Customer Expectations:** evah segats lla ta snoitatcepxe remotsuC dna ssecca ot reisae si ti ,smroftalp enilno eht ot euD .elbaveihca erom emoceb snoitome rieht ot gnidrocca segassem ro snoitca laiceps gnipoleved ,stegrat ecnivnoc .lennuf esahcrup eht no pets ro
4) **Focussed Marketing Efforts:** neewteb pihsnoitaler eht sevorpmi dna sezimitpo tl tegrat no desab snoisiced gnitekram retteb ekam ot spleh ,sremusnoc dna sdnarb .lortnoc rednu niahc ylppus eht lla evah ot swolla dna snoitcaer

COVID-19 crisis

- he COVID-19 crisis has highlighted the need for significant support and investments on digital transformation and effective digital governance across all countries in the region, particularly to ensure the continuity and delivery of core government functions.
- In the past months, business continuity was severely hampered, and many government officials and staff had to continue working in their physical offices not only due to infrastructure issues, but most importantly due to the presence and location of data and information in offline data files and paper-based systems.

- the impact of the COVID-19 pandemic on digital transformation in various industries:



Retail

- The pandemic has led to an increase in online shopping, requiring retailers to adapt their online presence and fulfillment processes.





Health care

- The healthcare industry has relied on digital technologies such as telemedicine, electronic health records, and data analytics to support decision-making and resource allocation.

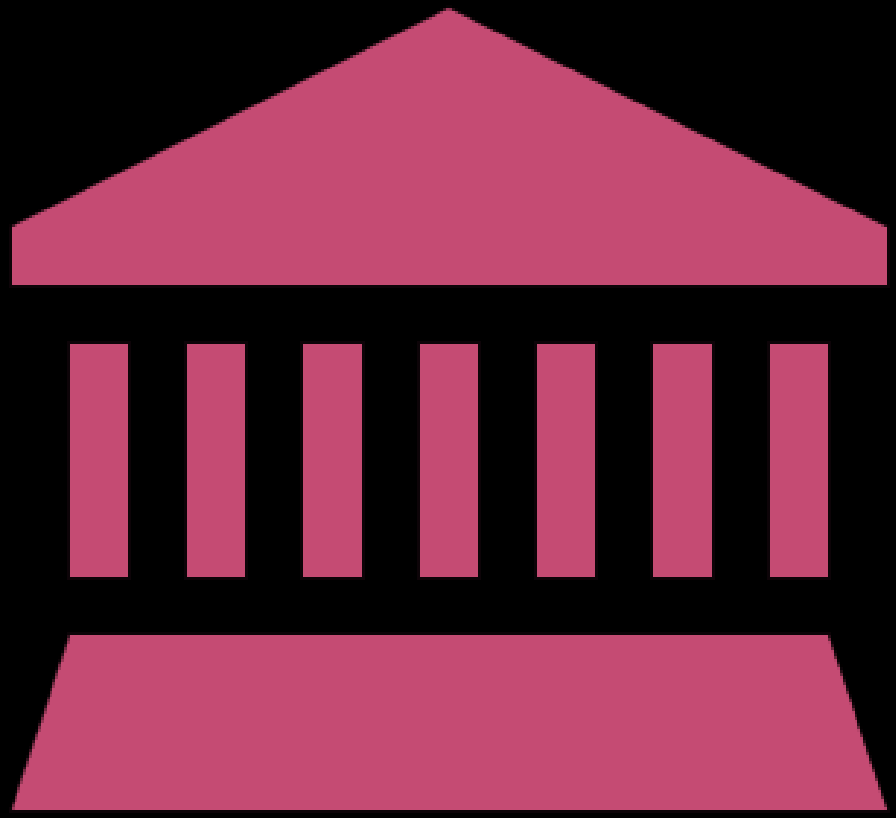
Education

- The closure of schools and universities due to the pandemic has led to a rapid shift to online education, requiring educators to adopt new technologies and approaches for remote teaching and learning.



Financial services

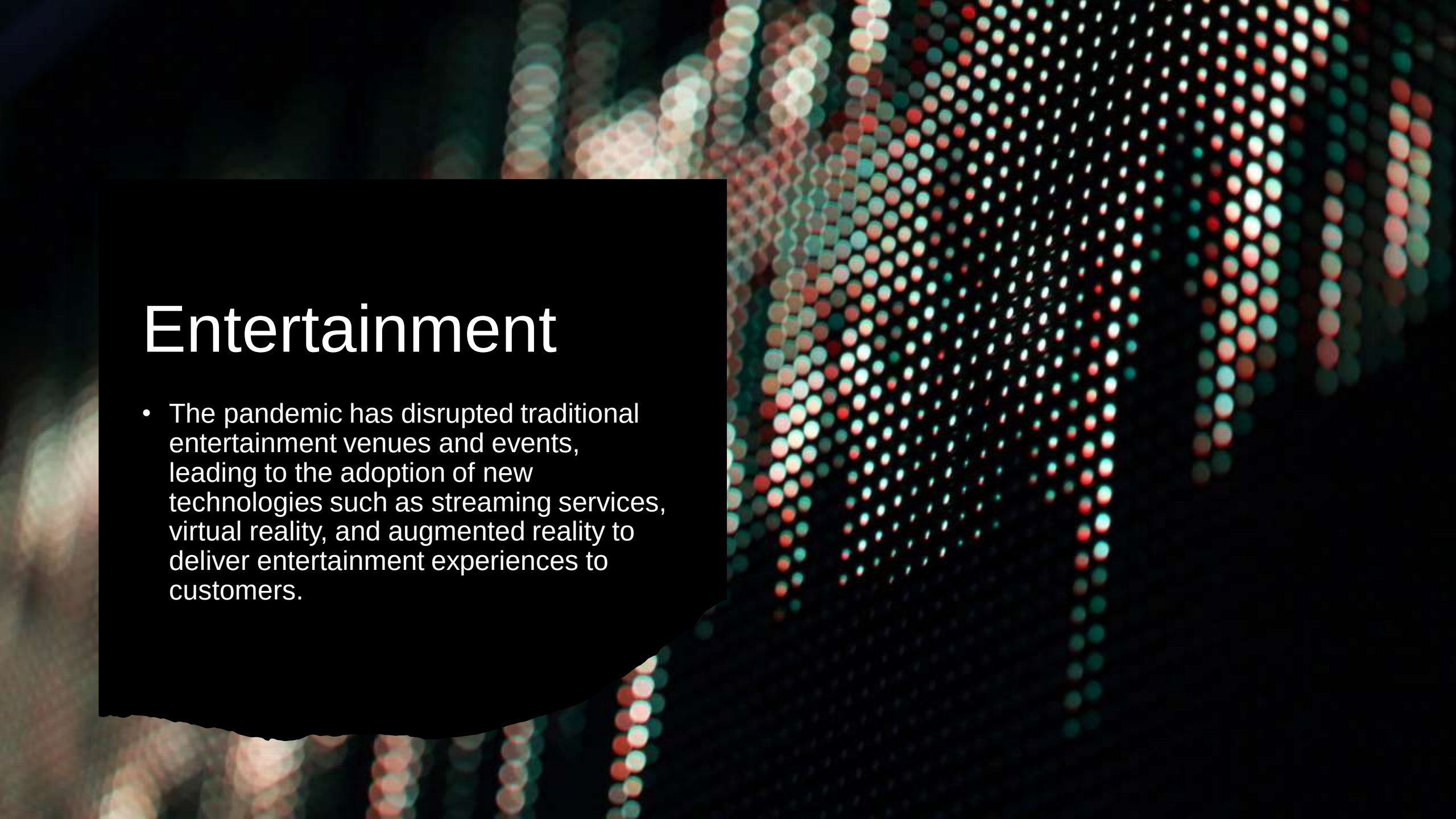
- The pandemic has led to an increase in the use of digital banking and financial services, as consumers have sought to minimize in-person contact and reduce their reliance on cash. This has required financial institutions to adapt their digital offerings and enhance their cybersecurity measures.





Manufacturing

- The pandemic has disrupted global supply chains and disrupted production, leading manufacturers to adopt new technologies such as robotics, 3D printing, and the internet of things (IoT) to improve efficiency and reduce risk



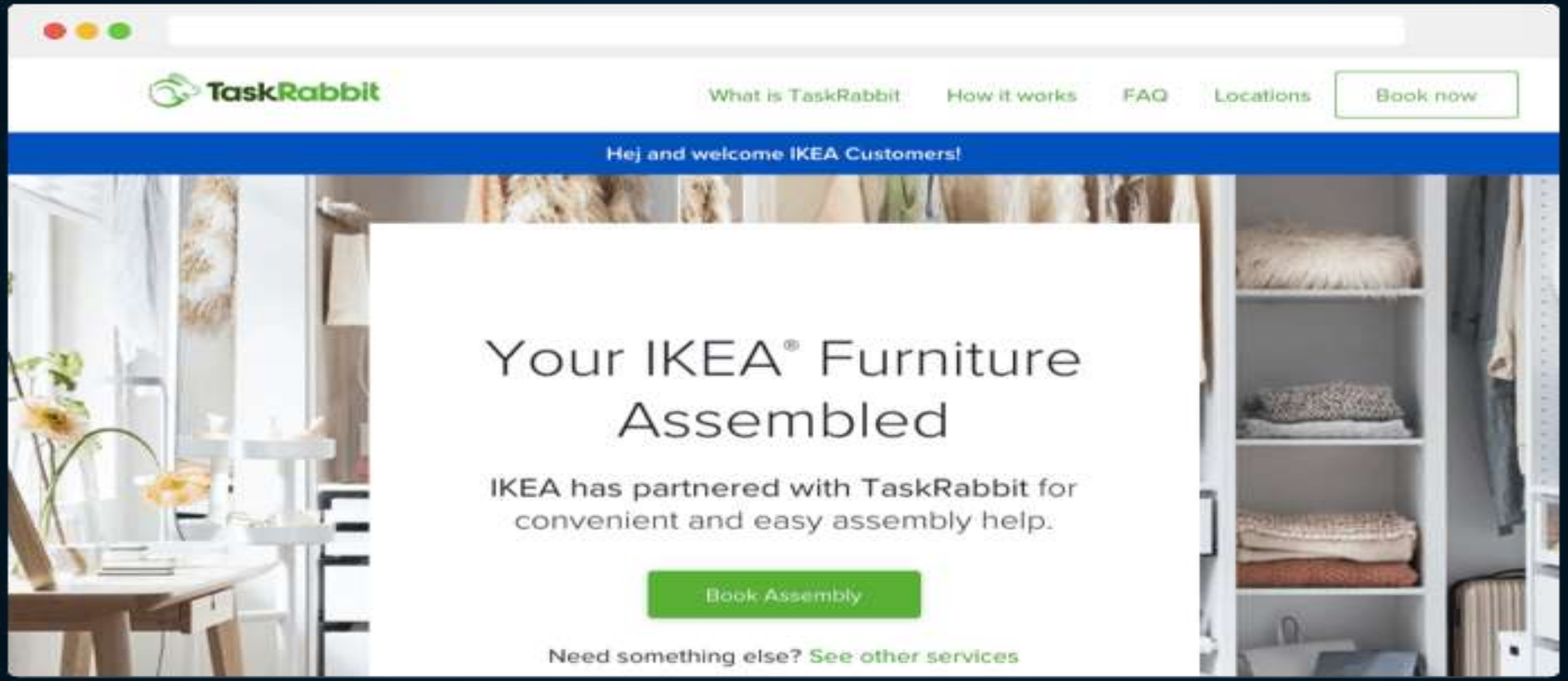
Entertainment

- The pandemic has disrupted traditional entertainment venues and events, leading to the adoption of new technologies such as streaming services, virtual reality, and augmented reality to deliver entertainment experiences to customers.

06

REAL-WORLD INDUSTRY EXAMPLES

IKEA


TaskRabbit
What is TaskRabbit
How it works
FAQ
Locations
Book now
Hej and welcome IKEA Customers!
Your IKEA® Furniture Assembled
IKEA has partnered with TaskRabbit for convenient and easy assembly help.
Book Assembly
Need something else? See other services

IKEA

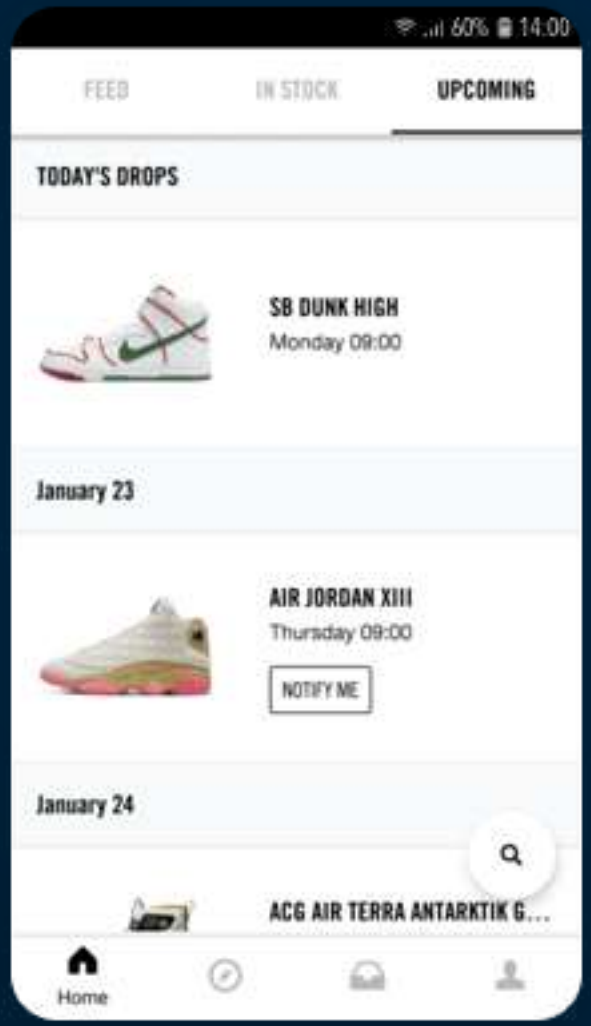
The Swedish furniture company is one of the digital transformation examples showing how DX can change a traditional organization. Thanks to new technologies, IKEA is changing the shopping experience of its customers for the better and optimizing its costs in the process.

In 2017, the company acquired TaskRabbit. This is a website that lets you search for people to help with the assembly or delivering furniture purchased at IKEA to your apartment. Thanks to this, customers who can't manage these tasks on their own are more likely to use the services of the company.

In addition, IKEA has decided to implement a smart home project. For the furniture industry tycoon this includes intelligent kitchen equipment and lighting. The company also uses Augmented Reality in the IKEA Place application to make it easier for customers to choose furniture and virtually "furnish" the apartment before making purchases in the store.
Another example of the digital transformation of IKEA is the Space10 project, which is a research center and exhibition center. It was there that the company presented the concept of autonomous cars as a residential and entertainment space. However, IKEA does not plan to produce cars alone, it only creates concepts for their interior.

NIKE


FEED
IN STOCK
UPCOMING
AIR JORDAN XIII
CHINESE NEW YEAR
NOTIFY ME
ACG AIR TERRA ANTARKTIK GORE-TEX
Home


FEED
IN STOCK
UPCOMING
TODAY'S DROPS
SB DUNK HIGH
Monday 09:00
January 23
AIR JORDAN XIII
Thursday 09:00
NOTIFY ME
January 24
ACG AIR TERRA ANTARKTIK G...
Home


DISCOVER
ELEVATED AIR
Jordan Brand's Best
Discover

NIKE

NIKE, a giant in the sports footwear industry, has started focusing on mobile devices. It uses advanced technologies based on recommendation algorithms and machine learning.

The company has created, among others, a mobile application that helps choose the best type of shoe based on a leg scan. By scanning the feet, the application creates a map of them based on 13 data points. What's more, thanks to the application, the company attains vital information for creating better footwear projects in the future.

An important project of the company is the NIKE + program, which rewards the most active loyalty program members. In Japan, where the solution was implemented, it led to a significant increase in sales. The NIKE SNKRS application was also successful, where the distinguished shoes recorded a 100% increase in sales.

NIKE's digital transformation shows that innovation can also lead to modernizing internal company operations. By digitizing 6,000 footwear materials, company design teams can work faster and more efficiently. This directly affects the speed of response to market needs.

The Swiss-based specialty coffee machine manufacturer and operating unit has deployed a cloud-based CRM system that provides customers with comprehensive access to shopping and customer service. Customers can access the company whether they are using the website, using a mobile device, or visiting a store. Having a single view of each of its customers – a 360-degree view – has enabled Nespresso to move into more markets and improve the sales.



Netflix. Founded as a mail-order DVD rental company in 1997, it has reinvented itself as an online video streaming service with personalized offers based on each customer's preferences.




amazon


CapitalOne


airbnb


Domino's
Pizza